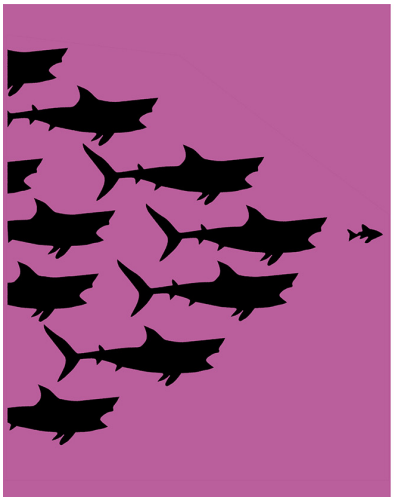


LESSONS FROM MATHEMATICAL THEORY ON THE STRUGGLE FOR LIFE

PREDATOR–PREY EQUATIONS

IN CONTEXT

KEY FIGURES

Alfred J. Lotka (1880–1949), **Vito Volterra** (1860–1940)

BEFORE

1798 British economist Thomas Malthus shows that the rate at which the population changes increases as the size of the population grows.

1871 In Lewis Carroll's novel *Through the Looking Glass*, the Red Queen tells Alice, "you have to run just to stay in the same place."

AFTER

1973 American biologist Leigh Van Valen proposes the Red Queen effect, which describes the constant "arms race" between predators and prey.

1989 The Arditi–Ginzburg equations offer another model of predator–prey dynamics by including the impact of the ratio between predator and prey.

The predator–prey equations are an early example of the application of mathematics to biology. Formulated in the 1920s by American mathematician Alfred J. Lotka and Italian mathematician and physicist Vito Volterra, the two equations—also known as the Lotka–Volterra equations—describe the way in which the population of a predator species and that of its prey fluctuate in relation to each other.